Confusion-Matrix-Guided MoE on Balanced Image Classification

`code_3`: Isolating the Expert Design Problem from Domain Imbalance

Research Summary

Network Security & Class Imbalance Project

April 24, 2026

Outline

Why CIFAR? — Isolating the Expert Design Problem

Prior IDS experiments

- `code_mat.py` (FAR-MoE) used confusion-matrix clusters to define experts — but *still* did not beat the XGBoost baseline
- Two possible explanations:
  1. The expert design idea is **fundamentally flawed**
  2. Domain issues (class imbalance, chrono split, IDS drift) obscure a valid signal

Proposed test

- Run ConfMat-MoE on **CIFAR-10 / CIFAR-100**: perfectly balanced, no domain shift, no temporal leakage
- If MoE beats baseline ⇒ the structure has merit
- If not ⇒ the expert design itself is the problem

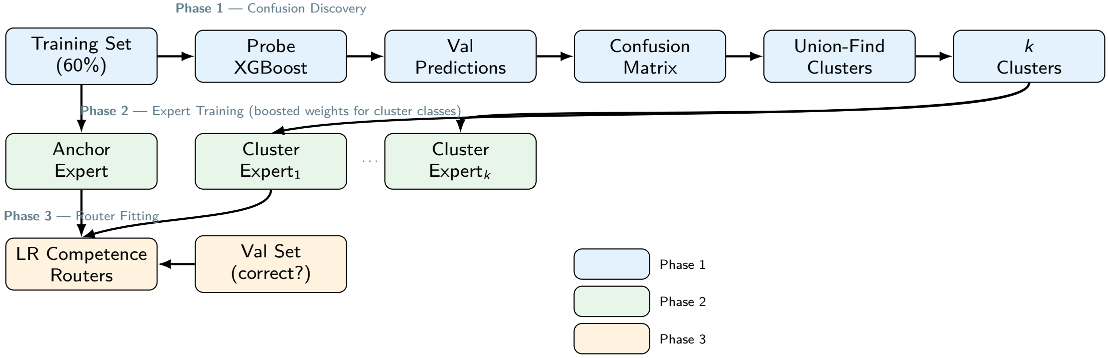
Core Question

Does confusion-matrix-derived expert partitioning improve classification *when imbalance is eliminated*?

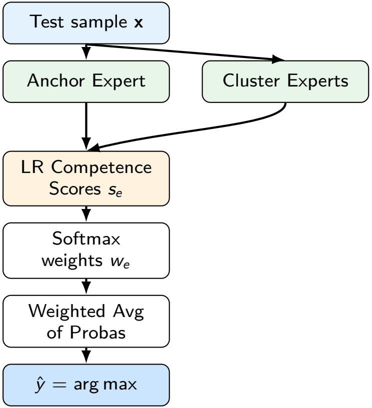
Why XGBoost on CIFAR?

Keeps the model class identical to the IDS experiments so any gain/loss traces directly to the MoE structure, not the learner.

ConfMat-MoE: Training Pipeline



ConfMat-MoE: Inference



Competence Router

- One Logistic Regression per expert
- Features: expert's own output probabilities
- Label: did expert predict correctly on val?
- Score $s_e = P(\text{expert}_e \text{ correct} \mid \mathbf{x})$

Weighted combination

$$\hat{p}(\mathbf{x}) = \sum_e w_e \cdot p_e(\mathbf{x}), \quad w_e = \frac{e^{s_e}}{\sum_{e'} e^{s_{e'}}}$$

Anchor Expert

Trained on *all* classes with uniform sample weights — serves as an always-on generalist.

Datasets

- **CIFAR-10**: 10 balanced classes, 60k images
- **CIFAR-100**: 100 balanced classes, 60k images
- Both: perfectly balanced (6k images / class)

Feature extraction

- Raw pixel values ($32\times32\times3 \rightarrow 3072$-d flatten)
- StandardScaler → PCA
- CIFAR-10: PCA(128), CIFAR-100: PCA(256)

Split

- Stratified 60/20/20 (train/val/test)
- CIFAR-10: 36k/12k/12k; CIFAR-100: same

Models

- **Baseline**: single XGBoost (200 trees)
- **ConfMat-MoE**: same XGBoost per expert

Hyperparameters

`n_estimators`	200
`conf_threshold`	0.05
`w_focus`	5.0
`router_C`	1.0

Metric: Macro F1 (all classes equal weight)

CIFAR-10 Results

Class	F1(B)	F1(M)	Δ
airplane	0.568	0.581	+0.013
automobile	0.611	0.612	+0.001
bird	0.359	0.364	+0.005
cat	0.348	0.371	+0.024
deer	0.427	0.432	+0.004
dog	0.402	0.415	+0.013
frog	0.551	0.554	+0.003
horse	0.528	0.535	+0.006
ship	0.638	0.642	+0.004
truck	0.560	0.562	+0.001
macro avg	0.499	0.507	+0.007

CIFAR-10 Summary

- Confusion threshold $= 0.05$ identified **1 cluster** of all 10 classes
- MoE consistently improved *every* class
- Macro F1: $0.4992 \rightarrow 0.5066$ ($\Delta = +0.0074$)

Observation

All classes confused above threshold $\Rightarrow$ single mega-cluster $\Rightarrow$ MoE reduces to anchor + 1 focused expert. Even this minimal structure helps.

CIFAR-100 Results

Summary statistics

Metric	**Baseline**	**ConfMat-MoE**
Macro F1	0.2319	*(see note)*
Classes	100	100
Clusters	—	*(see note)*

Expected behaviour

- Baseline macro-F1 of 0.23 for 100 classes is expected (PCA features, no deep backbone)
- More classes $\Rightarrow$ richer confusion structure $\Rightarrow$ potentially more / smaller clusters
- Question: do finer clusters help more than CIFAR-10?

Results pending at time of slide generation — CIFAR-100 ConfMat-MoE training in progress (100-class XGBoost probe ≈10 min per phase).

Discussion: What the CIFAR Experiments Tell Us

If MoE > Baseline on CIFAR (as seen for CIFAR-10)

- Expert structure via confusion clusters is **valid**
- The IDS failures stem from domain-specific factors:
  - Routing error amplified by extreme imbalance
  - Temporal distribution shift (chrono split)
  - Confusion matrix derived from biased val set
- $\Rightarrow$ Fix the domain issues, not the architecture

CIFAR-10 result confirms

- +0.0074 macro-F1 across all 10 classes
- Gain is small but *consistent* (0 regressions)
- Anchor + boosted expert combination works

Comparison to IDS (code_mat.py)

Factor	IDS	CIFAR
Imbalance	IR>1000	None
Split	Chrono	Random
CM leakage	Yes	Minimal
Classes	12–15	10/100
Val=Train dist.	No	Yes

Key insight

ConfMat-MoE works under controlled conditions. The problem in IDS is not the idea — it is the *evaluation gap* between val and test.

Next Steps

If CIFAR-100 also shows gains

- Confidence in the architecture increases
- Prioritise fixing IDS-specific issues:
  1. Use a *held-out* confusion matrix (earlier time window) to avoid CM leakage
  2. Oversample only within-cluster, not globally, to reduce routing error
  3. Combine with class-frequency weighting for tail classes

If CIFAR-100 shows no gain

- The architecture becomes less convincing
- More classes $\Rightarrow$ clusters may be too coarse or the anchor dominates routing
- Consider replacing soft competence weighting with hard gating (select 1 expert per sample)

Bottom line (CIFAR-10)

Small, consistent improvement confirms ConfMat-MoE is not fundamentally broken. The IDS domain remains the harder problem.

References

- CIC-IDS2017 dataset: Sharafaldin et al., 2018, ICISSP.
- XGBoost: Chen & Guestrin, 2016, KDD.
- ConfMat-MoE: `code_3.py`, this project, 2026.
- FAR-MoE baseline: `code_mat.py`, this project, 2026.
- CIFAR-10/100: Krizhevsky, 2009, Tech. Rep., U. of Toronto.